

Customer Shopping Behavior Analysis

Transactional insights from **3,900 purchases** across demographics, categories, and subscription behavior



Dataset at a Glance

3,900

Transactions

18

Columns

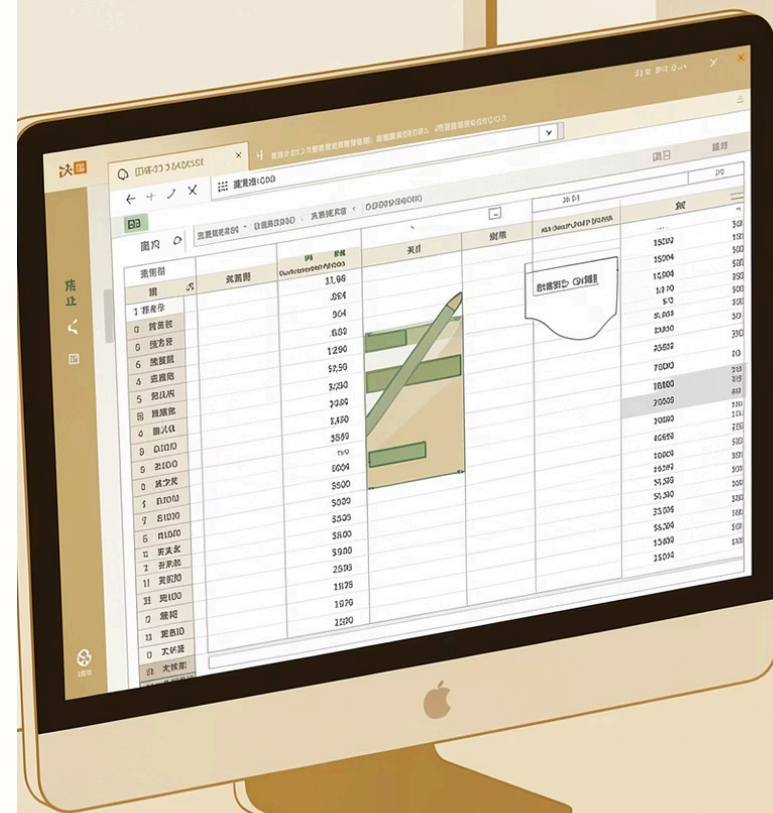
37

Missing Values

Review Rating only

\$59.76

Avg Purchase



Data Preparation in Python



The image shows a code editor window with Python code for data preparation. The code includes comments in French and uses the pandas library. To the right of the code is a table with data.

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→ Pandbas	23930	1-1	5317
Pandbas	23330	1:0	
→ Pandbas	58590	1.8	
Pandbas	24554	10	Scored

→ Clean & Standardize

Snake_case columns, median imputation for Review Rating

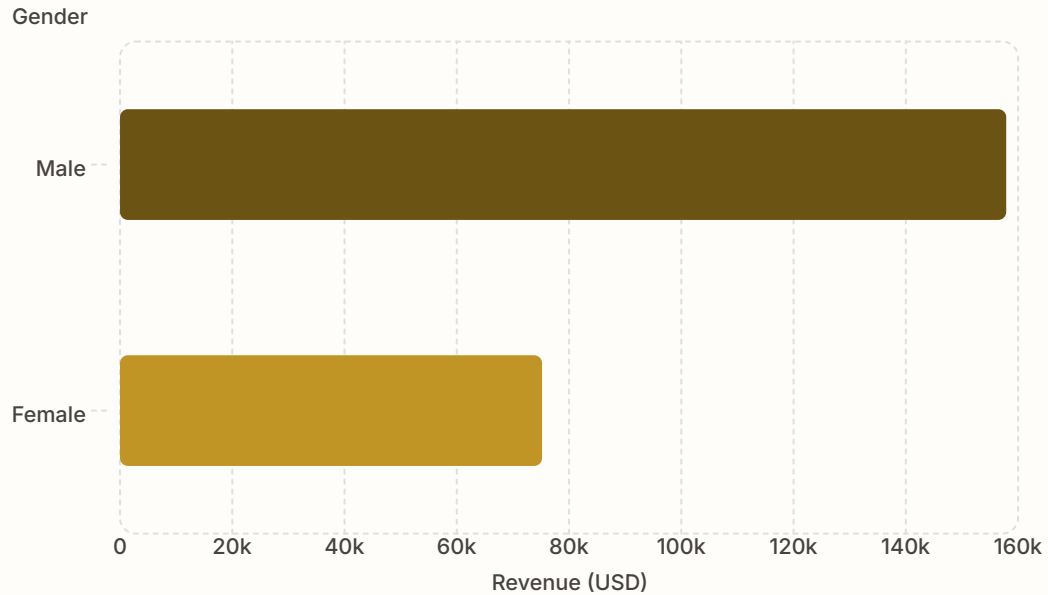
→ Feature Engineering

Created age_group and purchase_frequency_days

→ Load to PostgreSQL

Cleaned DataFrame loaded for SQL analysis

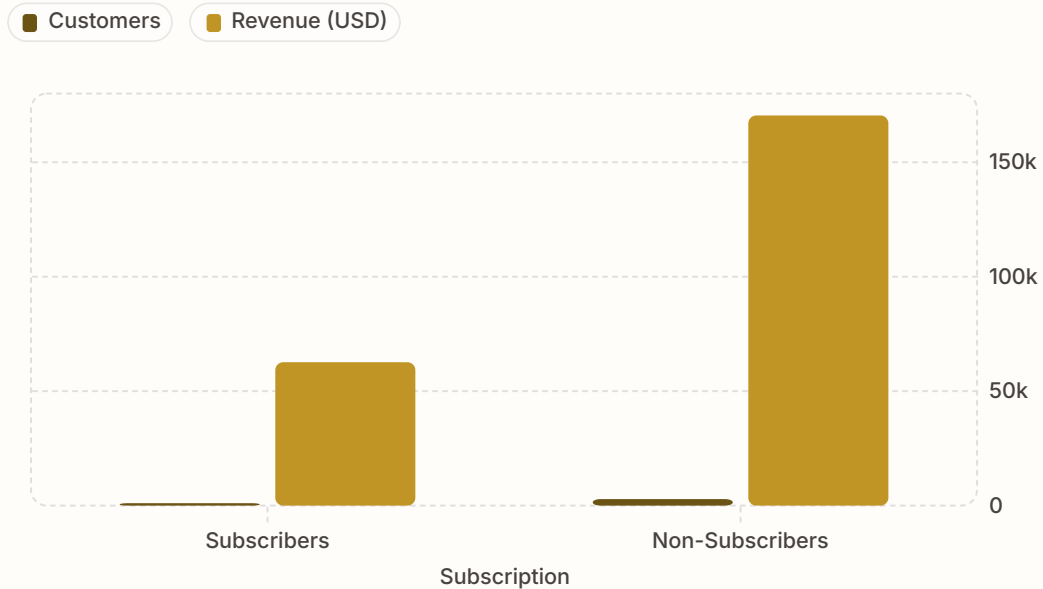
Revenue by Gender



Key Finding

Male customers generated **2.1x more revenue** than female customers, despite females representing a significant portion of buyers.

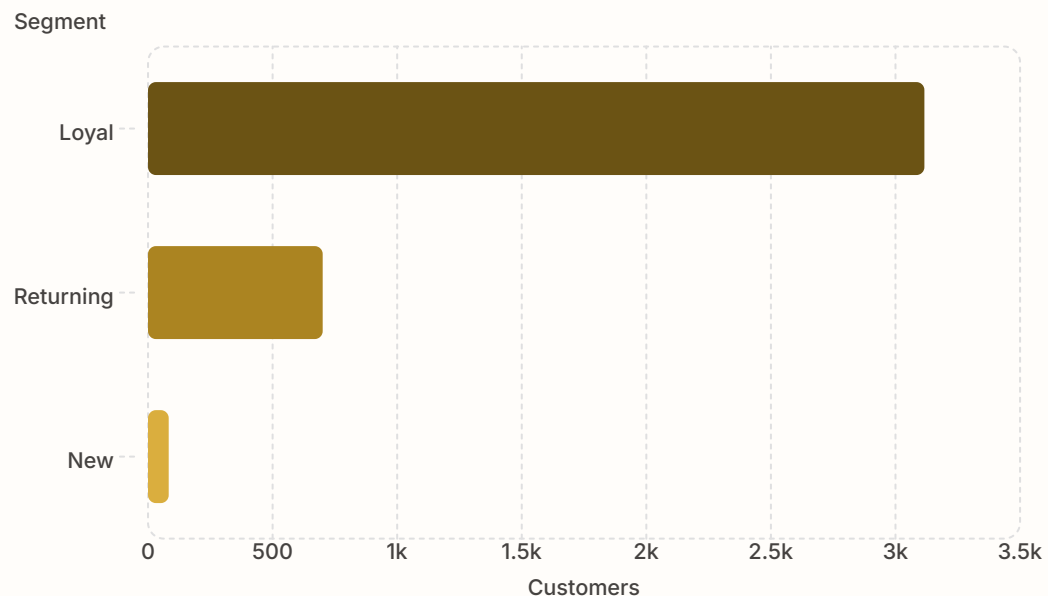
Subscribers vs. Non-Subscribers



Insight

Only **27%** of customers subscribe, yet average spend is nearly identical (~\$59). Large untapped opportunity.

Customer Segmentation



Segment Breakdown

Loyal — 80%

3,116 customers

Returning — 18%

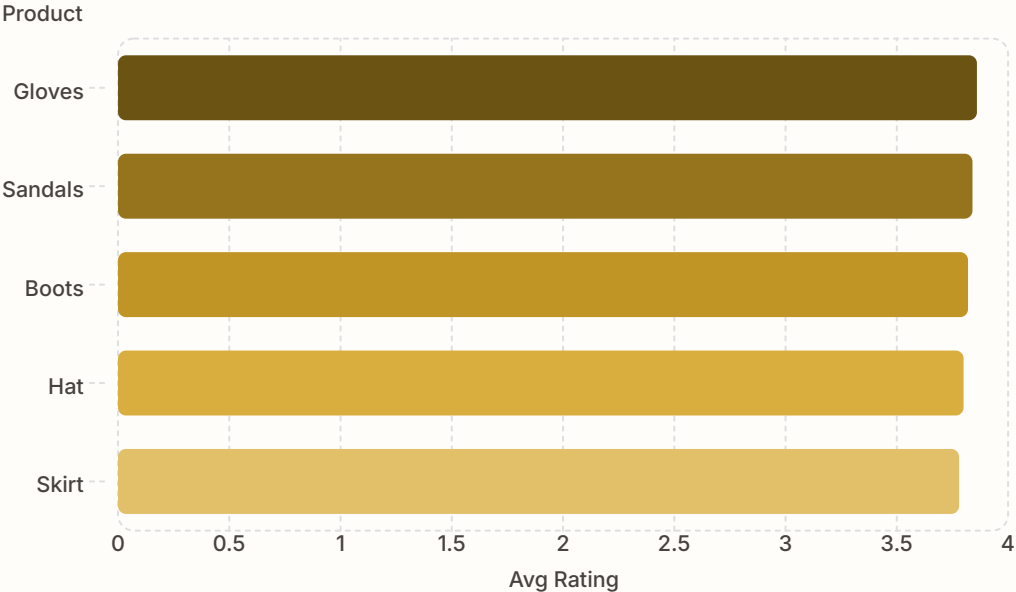
701 customers

New — 2%

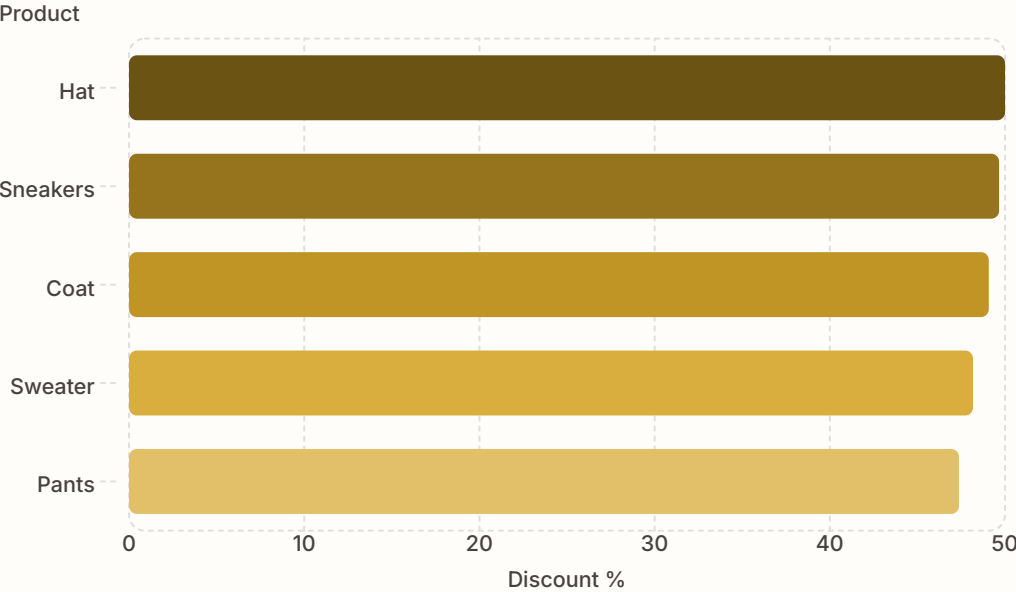
83 customers

Top Products & Discount Patterns

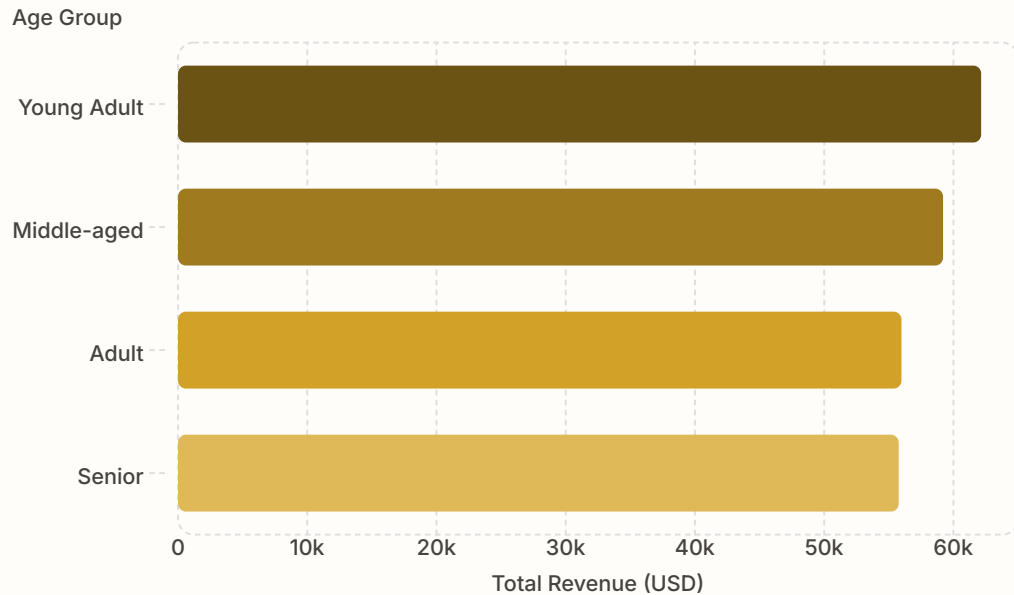
Top 5 by Review Rating



Highest Discount Rates



Revenue by Age Group



Observation

Revenue is **remarkably balanced** across age groups — Young Adults lead slightly, but all segments contribute meaningfully.

Power BI Dashboard

KPIs

3.9K customers · \$59.76 avg
purchase · 3.75 avg rating

Filters

Subscription, Gender, Category,
Shipping Type

Visuals

Revenue & Sales by Category and Age Group



Business Recommendations



Boost Subscriptions

Promote exclusive subscriber benefits



Loyalty Programs

Reward repeat buyers toward Loyal segment



Review Discounts

Balance sales lift with margin control



Targeted Marketing

Focus on high-revenue age groups & express shippers